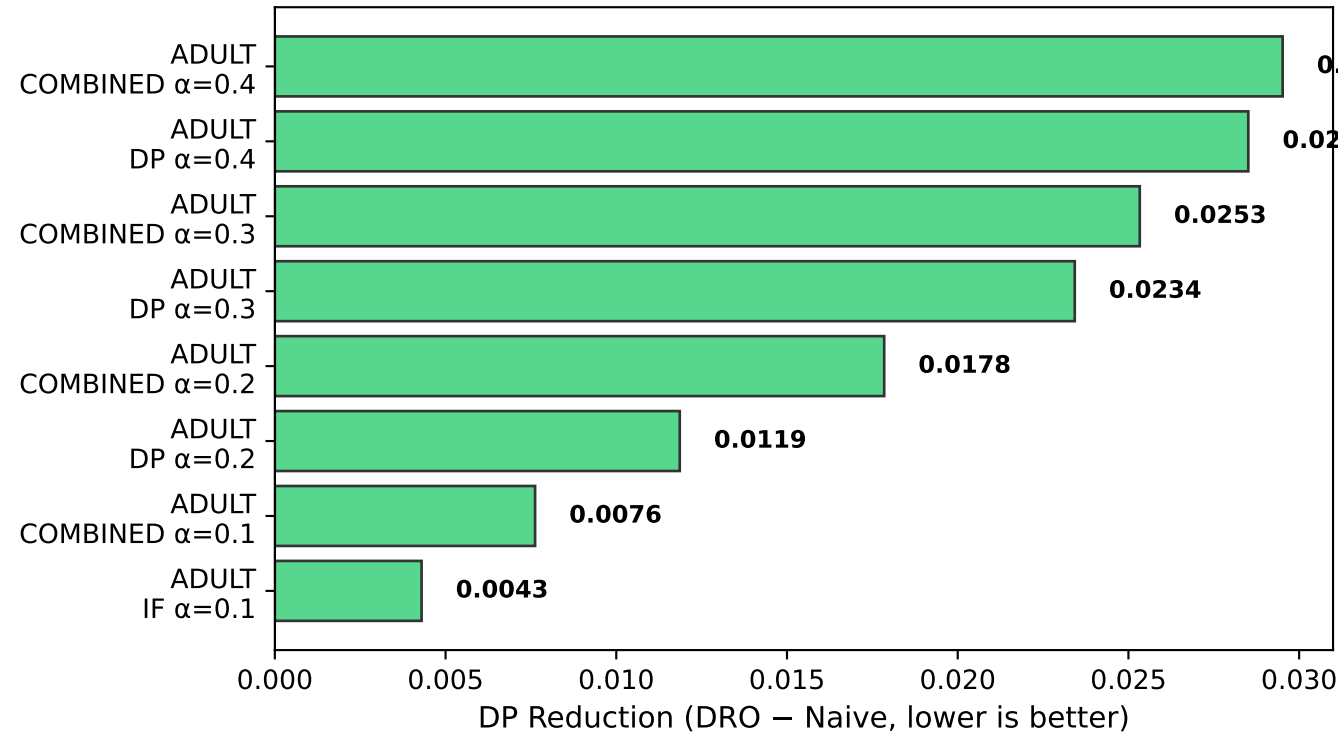
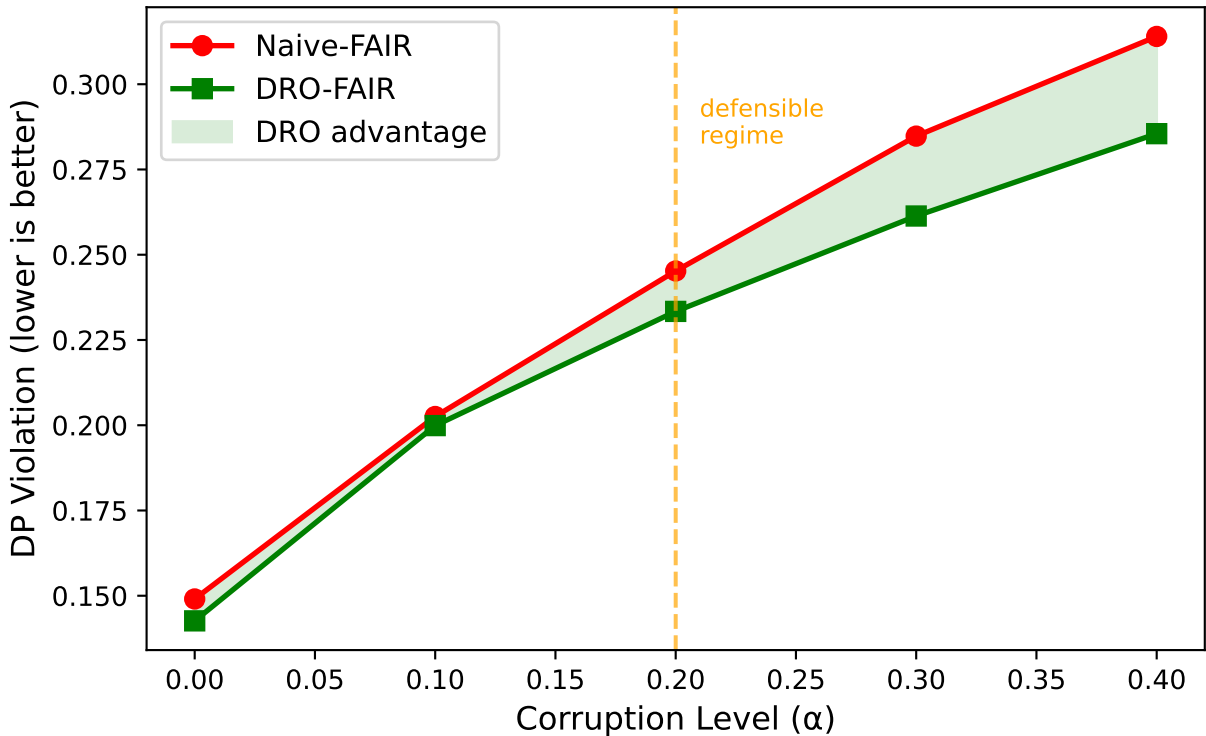


Kuldeep Meeting — Jun 30 2026 | DRO-FAIR: Adversarial Robustness (tau=1 fixed)

DRO vs Naive: Statistically Significant DP Wins
(Wilcoxon $p < 0.05$, $n=6$ seeds, $\tau=1$ fixed)



Adult — DP Attack: DRO vs Naive
($\tau=1$ fixed, $K_{\text{inner}}=10$, 6 seeds)



KEY FINDINGS (Jun 30 2026) — $\tau=1$ fixed, $K_{\text{inner}}=10$, 6 seeds, FairnessTargetedPGD

HEADLINE: DRO-FAIR beats Naive-FAIR on DP at every alpha (Adult, $n=6$, all $p < 0.05$)
PREVIOUS ARTIFACT: $\tau=100$ schedule made DRO look fragile — fully resolved with $\tau=1$ fixed

Defensible regime: $\alpha \leq 0.2$ (acc ≥ 0.76 , beats constant-label predictor 0.752)
At $\alpha \geq 0.3$: 30-40% label-corruption ceiling — constant predictor dominates (expected)

Adult DP attack: DRO advantage grows with alpha (delta 0.006 $\rightarrow$ 0.029, all $p=0.016$)

Adult Combined: DRO wins all alphas (all $p < 0.05$)

Credit: DRO wins DP at all alphas ($n=6$ for $\alpha \leq 0.2$)

LSAC: pending (completing autonomously via watchdog)